OBJECT ORIENTED SOFTWARE ENGINEERING

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Architectural Styles

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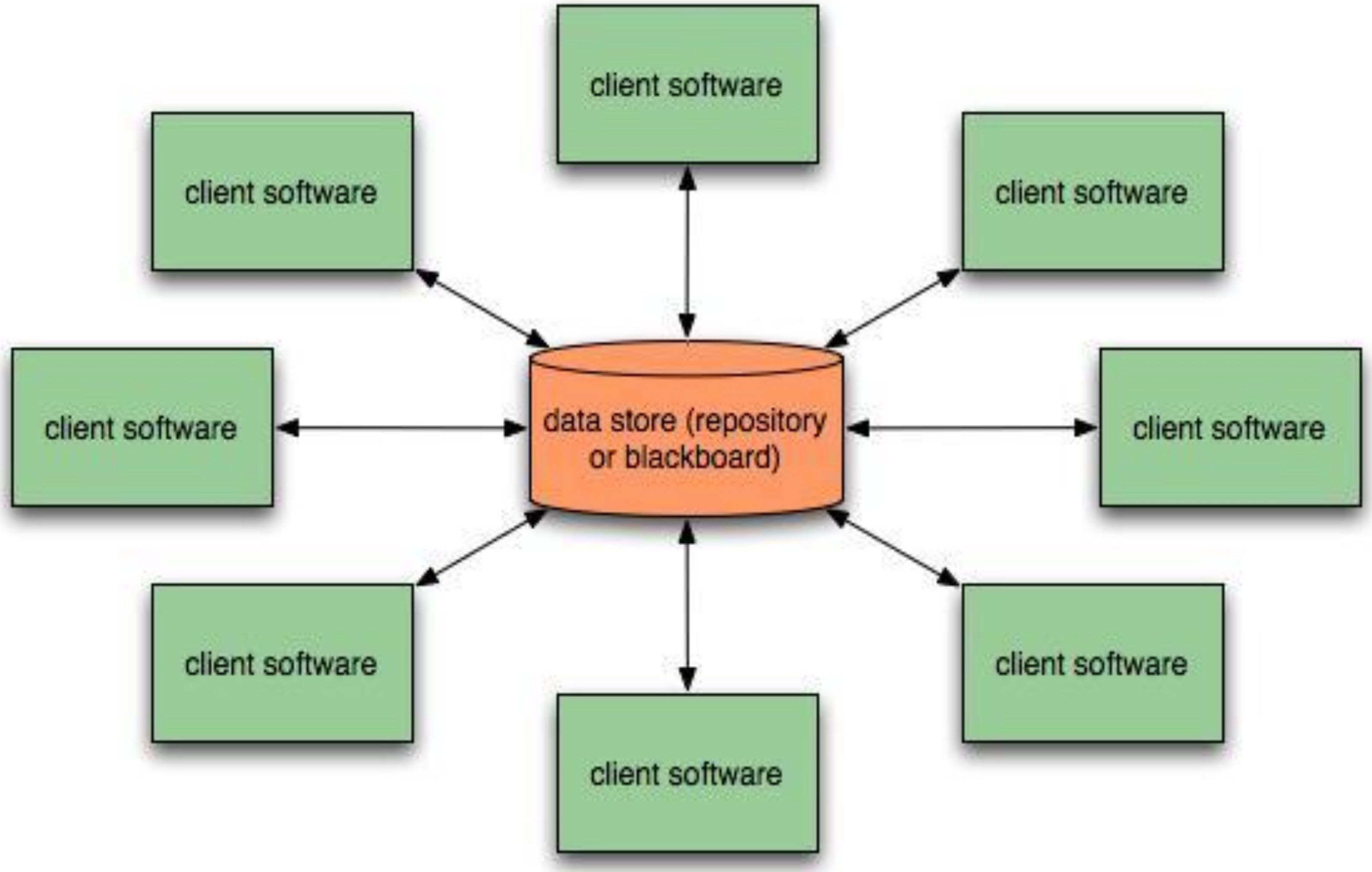
Architectural Styles

1. **Data-centered architectures**
2. **Data flow architectures**
3. **Call and return architectures**
4. **Object-oriented architectures**
5. **Layered architectures**

1. Data Centered Architecture



1. Data Center Architecture Style

- Has the goal of **integrating the data**
- A data store (e.g. a file or database) resides at the center of this architecture and is accessed frequently by other components that **update, add, delete or modify data** within the store.(shown in fig.)
- Client software accesses a **central repository**.
- In some cases data **repository is passive**. i.e., client software accesses the data independent of any changes to the data.
- A variation on this approach transforms the repository into a "**blackboard**", that **sends notifications to client software when data of interest to the client changes**

1. Data Center Architecture Style

- Data-centered architectures promote ***integrability***
  - i.e., existing components can be changed and new client components added to the architecture without concern about other clients (since the client component operate independently).

- In addition, data can be passed among clients using blackboard mechanism.

- Client components independently execute processes

THANK YOU

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